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Organic Evolution

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Organic Evolution

- Evolution is descent with modification
- Organic Evolution is the origin of new species of animals and plants from ancestors who previously lived on the earth.
- There are two different types of development, viz., ontogeny, and phylogeny.
- "Ontogeny" is primarily concerned with the history of development of individual organisms.
- "Phylogeny" deals with the evolution of a genetically related group of organisms, in contrast to the development of the individual organism.

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Pre Darwinian Phase

- In the fourth century B.C., Theophrastus, a Greek, has written a book on seed germination.
- Some early Greeks believed that heredity among humans was a question of which sex dominated in the sex act.
- **Preformation** - Every organism must therefore contain in its reproductive organs an infinite series representing all of its future descendants".
- Ovists believed the female possessed this future of the series, and the spermists insisted it was the male.

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Pre Darwinian Phase

- Linnaeus suggested a two-layered theory, which held that "the outer layer including the vascular system is derived from the father, the inner layer including the nervous system comes from the mother".
- That plants lacked sexuality was the conventional wisdom of the day.
- End of the seventeenth century - the presence of sex organs in plants demonstrated by Rudolph Camerarius.
- Hybridization experiments, finally culminated in Universal Law of Genetics by Mendel

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Theory of Inheritance of Acquired Characters

- Jean Baptiste de Lamarck
- ***Modifications which the organism acquires in adaptation to the environments which it meets during its life time are automatically handed down to its descendants and so become part of heredity.***
- "Philosphie Zoologique" - 1809

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Lamarckism

Four propositions

1. Living organisms and their component parts tend continually to increase in size.
2. Production of a new organ result from new need and from the new movement which this need starts and maintains.
3. If an organ is used constantly, it will tend to become highly developed, whereas disuse results in degeneration.
4. Modification produced by the above principles during the life time of an individual will be inherited by its offspring.

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Lamarckism

- Lamarck believed that organic changes seen in animals are a result of the influence of environment on gradual changes of species due to their tendency to become more and more perfect.
- Ex. Long neck of Giraffe, blindness in moles, webbed feet of the duck
- Lamarckian theory was simple and it had some appeal as it provided a way in which changes in organisms could come about.
- It was the first complete comprehensive theory that was offered.
- Thus, Lamarckian theory enjoyed popular acceptance for about 70 years, because it was exemplified by many common examples.

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Critical Analysis of Lamarckism

- All four propositions fail the test of empirical validity
- Experiments disproved Lamarckism – August Weismann

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Darwinism

- Charles Darwin and Alfred Russell Wallace
- 1859 – "Origin of Species & the Preservation of Favored Races in the Struggle for Life"
- **Theory of Natural Selection**
- ***"The change in species by the survival of an organismal type exhibiting a natural variation, that gives it an adaptive advantage in an environment, thus leading to new environment equilibrium, is evolution by natural selection".***

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Darwinism

Five Postulates

1. Universal occurrence of variation
2. Excessive natural rate of multiplication (prodigality)
3. Struggle for existence
4. Survival of the fittest
5. Inheritance of variation

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Critical Analysis

- A veritable storm broke out with the publication of this book
- Was opposed primarily on ethical and religious grounds
- "Survival of the Fittest" – erroneously interpreted by people as "tooth and claw", "kill or be killed" kind of struggle
- Led to rationalizations likes "every man for himself" in social and economic lives – **materialistic doctrine**
- Greatest weakness – did not account for inheritance – pangenesis hypothesis – disproved by Galton

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Critical Analysis

1. Talks about survival of the fittest but not arrival of the fittest
2. Overspecialization in certain organisms – cannot rationalize with natural selection
3. Natural selection did not account for degeneration
4. Dilution
5. Indirectly accepted Lamarckism – inheritance of acquired characters

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Post Darwinian Phase

- Between later 19th and early 20th century – a number of important ideas were forwarded to explain inheritance
- Discovery of **Mendel's laws of inheritance** by Correns and Tschemark
- Wagner – role of **geographical isolation** in speciation
- Hugo de Vreis – **Mutation theory**
- Research in **population genetics** proved that Darwin's struggle for existence between individuals is not a very significant force in evolution

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Post Darwinian Phase

- Unit of evolution is recognized as POPULATION and NOT INDIVIDUAL as Darwin believed
- **What is a Population? A group of individuals living within a circumscribed area and interbreeding**
- The next larger unit of population in nature is the **species**, composed of a series of inter-breeding demes.

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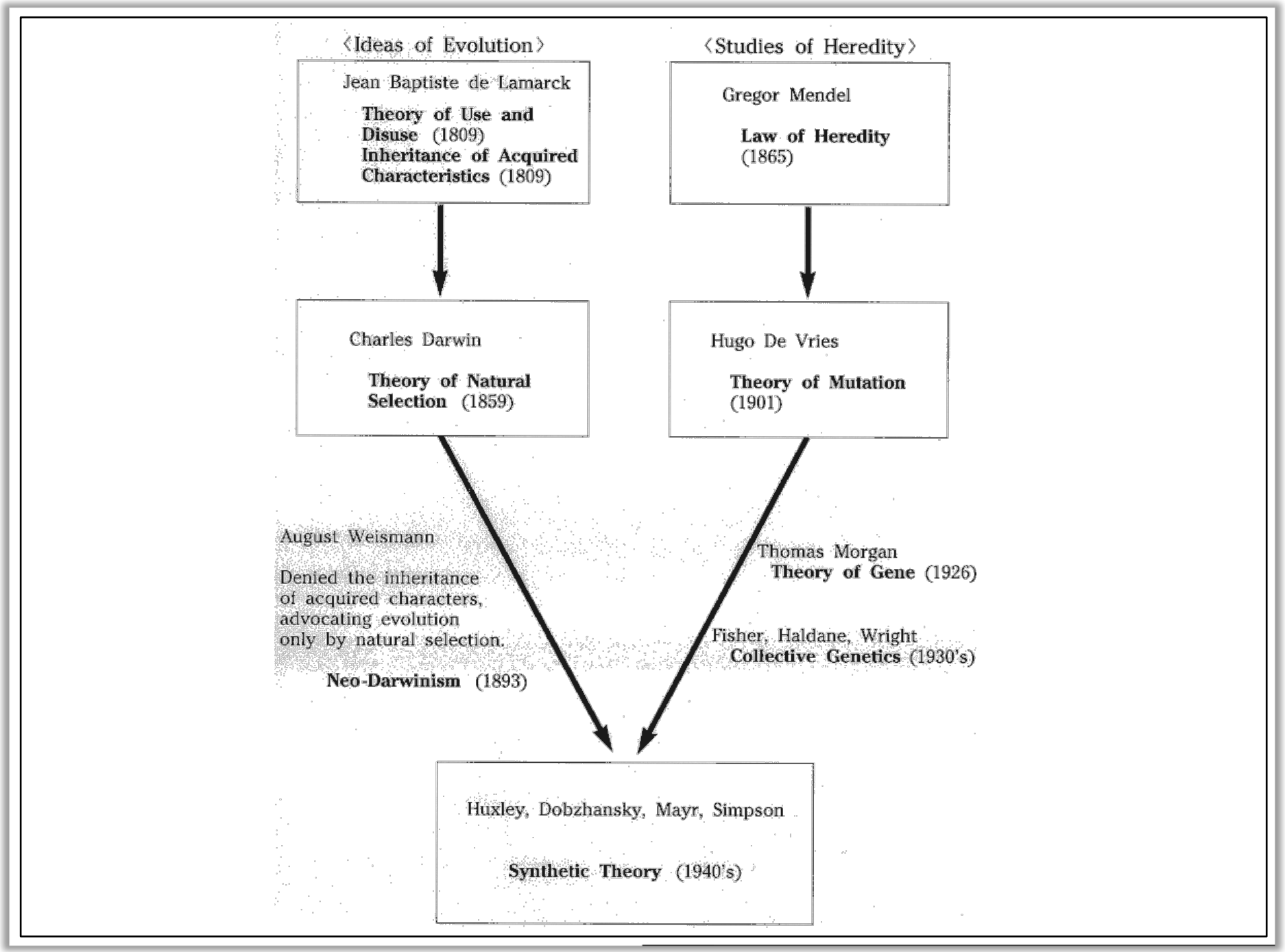
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The Modern Synthesis

- All the modern understandings in cytology, genetics, cytogenetics, population genetics, and evolution gave a way for the formulation of a coherent theory called **"Modern Synthesis"** around 1930s'
- S. Wright, H.J. Muller, Th. Dobzhansky, R.B. Goldschmidt, J.S. Huxley, R.A. Fisher, J.B.S. Haldane, Ernst Mayr and G.L. Stebbins, Simpson
- **It is a "synthesis" of "mutationist" and "selectionist" views of evolution**

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What is Evolution?

Evolution is a two stage process

1. Production and redistribution of variation
2. Action of natural selection on variation

- Evolution is defined as **change in the allelic frequencies in a gene pool of a given population from one generation to the next**
- The gene pool is the sum total of all the allelic genes in a population

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Unit of Evolution

- Alleles – variants of a gene
- Variation in population – because of the presence of alleles and their inheritance
- **The allele frequencies refer to only the whole group of individuals i.e. populations.**
- **Individuals do not have allelic frequencies;** they have the genes or the combination of these genes – genetic composition is fixed
- **Therefore, an individual cannot evolve; only a group of individuals – a population – can evolve over time.**

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Macro & Micro Evolution

- Evolution is an incredibly common phenomenon and may occur between every generation
- Over a period of time, the relative proportions of alleles in the population will change, some may increase, and some decrease and still others may remain the same.
- Over a short run of just a few generations, such changes in inherited traits may be only very small, but over a very long period of time, they may produce spectacular kind of adaptations and whole new variety of life
- Microevolutionary and macroevolutionary changes

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Factors producing variations

1. Mutations
2. Migrations
3. Genetic Recombination
4. Genetic Drift / Random Drift / Sewall Wright Effect

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Role of Natural Selection

- **What enables populations to adapt to environment? –** Natural Selection
- Genetic variations are created by mutations, migrations etc.
- Some of these variations influence **reproductive success** (number of offspring successfully raised)
- If, as a result of genetic variation, some individuals contribute more offspring to the next generation than the others, it is the role of natural selection
- Thus, natural selection is **differential net reproductive success**

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Role of Natural Selection

- **How do populations adapt?**
- A result of natural selection is a change in allele frequency relative to specific environmental factors
- **Such a functional shift in allele frequencies is called adaptation**
- **Unit of selection is individual**: It is individuals who reproduce or do not reproduce and who continually attempt to maximize their own reproductive success.
- **Extinction:** If the total reproductive success of all members of a population continuously falls below replacement value (where more individuals die than are born in a generation) **the population will become extinct.**

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Levels of Natural Selection

- Directional Selection – When environment is slowly and gradually changing in one direction
- Stabilizing Selection – When environment is stable
- Diversifying Selection – Extreme selection when drastic changes occur in environment

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Evolution at Species Level

- A species is defined as a group of interbreeding organisms that are reproductively isolated and, therefore, cannot successfully interbreed with other groups (species).
- A species is composed of subunits that are breeding communities, which we have already defined as populations.
- All members of a species can potentially interbreed, and some degree of interbreeding (migration) is theoretically possible between all populations of that species.

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Evolution at Species Level

- The net result of all forces of evolution acting on all populations determines the fate of the species as a whole.
- If sustained over a long period of time, gradual changes in allele frequencies between member populations can eventually lead to sufficient genetic differences, so that fertile reproduction is no longer possible.
- We then may recognize a new form of life having arisen from one species "splitting" and producing new species, a process called **speciation.**

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Isolation & Speciation

- A number of isolation mechanisms may also prevent organisms from interbreeding amongst each other
  - Geographic isolation
  - Behavioral isolation
  - Mechanical isolation
- Reproductive isolation is thus an important force that maintains biodiversity
- It also contributes to formation of new species by preventing interbreeding of two populations of the same species

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Summary

- Variation
  - Mutations
  - Recombinations
  - Migrations
  - Drift
- Natural Selection and Isolation act on variations
- Speciation and extinction

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Basic Concepts in Evolutionary Biology

Convergence

- Similar evolutionary development in **unrelated organisms or those that differ in direct ancestry**
- Could be similar characteristics or adaptations
- **Ex.** Flying habits of humming bird and humming moth – common search for nectar in flowers – hovering over flower
- Convergence ordinarily applies to one or two characteristics and not the overall makeup of organisms
- Only rods in retina – deep sea fish, bats, some lizards, lemurs

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Basic Concepts in Evolutionary Biology

Parallelism

- If convergence is similar evolutionary behavior in organisms which are unrelated in ancestry…
- Parallelism is similar adaptations in organisms which **share common ancestry**
- The term is usually applied to two species of organisms that were similar in origin, and that remained similar as they evolved into two different species.
- Ex. Old World Monkeys & New World Monkeys – separated 35 million years ago from a prosimian ancestor

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Basic Concepts in Evolutionary Biology

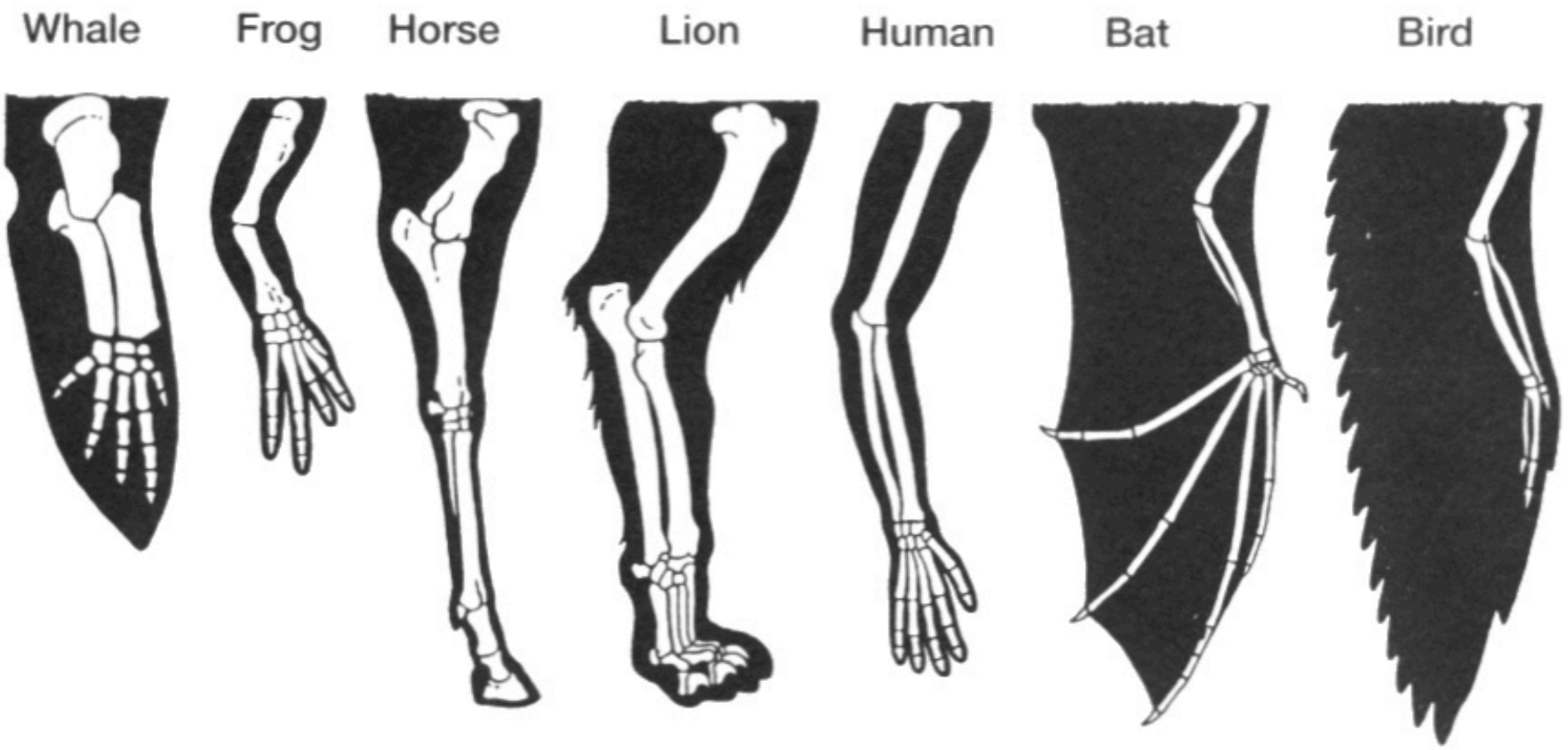
Homology & Analogy

- Organs which are similar in origin and not function are **homologous**
- **Ex. Forearms of vertebrates –** Forelimbs of vertebrates
- Organs which are similar in function and not origin are **analogous**
- **Ex. Wings of birds and wings of insects**
- **Serial Homology –** two structures within the same organism
- **Ex. Hands and Legs** in man

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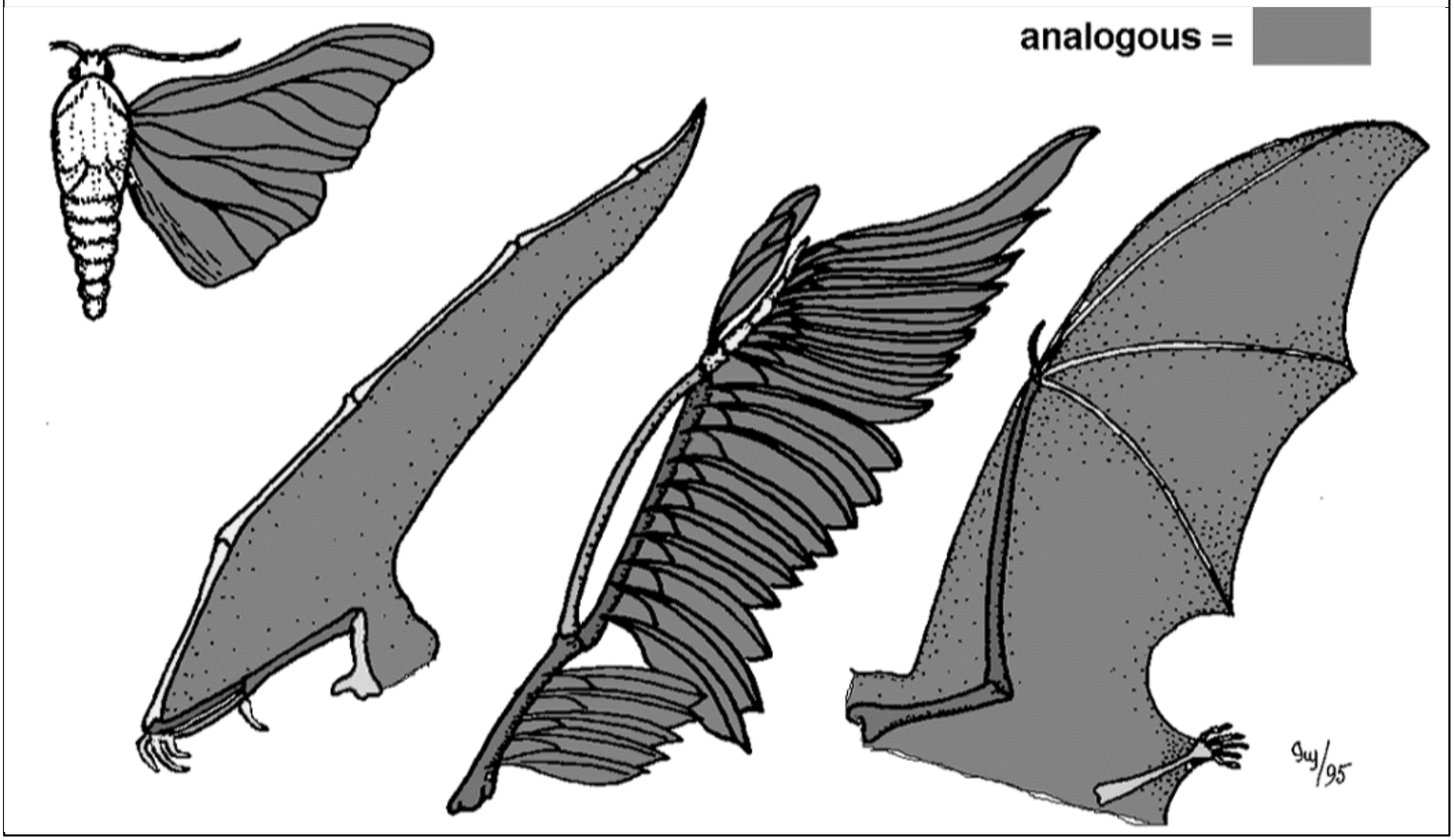
Homology



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Analogy



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Basic Concepts in Evolutionary Biology

Adaptive Radiation

- The evolutionary spread and differentiation of the descendants of one type of animal, of whatever level of classification
- Opposite of convergence and parallelism where different organisms arrive at common adaptation, radiation refers to the way a species evolves into progressively dissimilar organisms
- Mammals – Arial, arboreal, cursorial, fossorial, aquatic etc.
- Ex. Primates exhibit radiation in body size, group size, brain size etc.

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Basic Concepts in Evolutionary Biology

- **Dollo's Law –** Doctrine of Irreversibility of Evolution – Whales?
- **Cope's Rul**e – Lamarck's First principle
- **Gause's Principle –** Principle of Competitive Exclusion – if two organisms occupy the same trophic level in the ecosystem, they try to reach equilibrium through different strategies reducing the other to point of extinction
- **Mosaic Evolution –** Evolution of parts and not wholes – Hominization – Bipedalization followed by Encephalization

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